SHQ 3D

3D DESIGN & 3D PRINTING SOLUTIONS

MECHATRONICS & 3D PRINTING PORTFOLIO

by: @ariqshaq

SHQ 3D

SERVICE OFFER

> 3D DESIGN

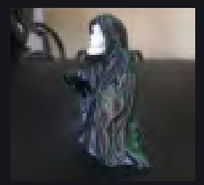
- We provide **3D Design by Request**, specializing in **Technical Design and Product Design** for your projects.
- Experienced in **Mechatronics Design**, where we provide a requested **Detailed Blueprint or Technical Drawing**, including component placement and selection.
- All designs are **Supported for Autodesk Softwares**.

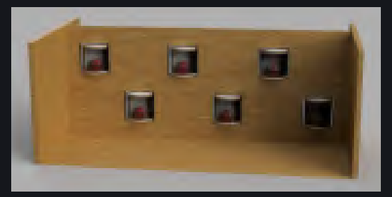
> 3D PRINTING

- We provide **FDM 3D Printing** supported for **PLA, PETG, and TPU Materials**.
- Your submitted designs can be **Consulted First or Adjusted by Us for Free**, so your designs fit well with the printing tolerance.
- **Guaranteed to Print Immediately** after the designs are submitted.
- We open to **Requests for Printing Settings**.

TRACK RECORD

- **Delivering Solutions Across Various Projects**, including lab research, campus projects, company requests, and competition clients.
- **Experienced with Over 3+ Years**, handling tight deadlines and complex specification requests.
- **Multiple Nationwide Recognitions** from national competitions to high-stakes projects.





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Contacts:

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SELECTED PROJECTS

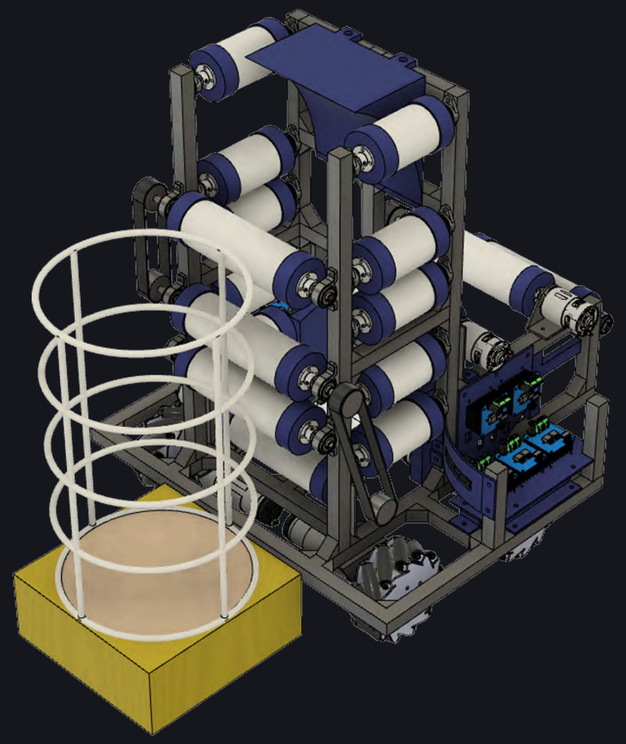
ABU ROBOCON 2024 ROBOT

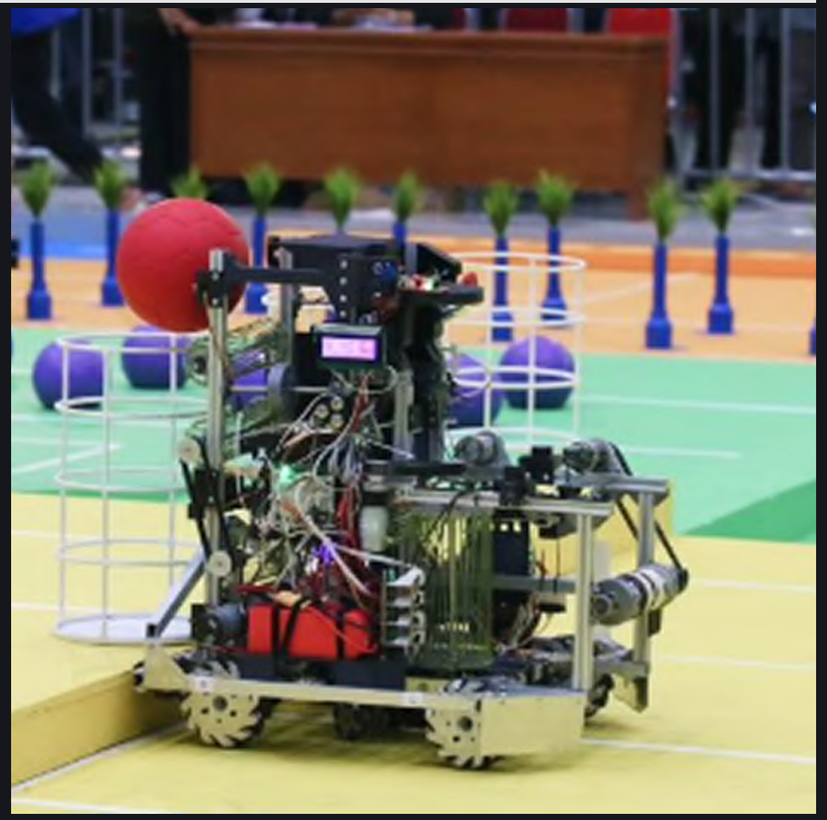
DESCRIPTION:

An Automatic Ball Sorting Robot, capable of picking and storing selected colored balls into silos.

TOOLS & SKILLS:

- 3D CAD
- 3D Printing
- Mechanism Design
- Grinding
- Drilling
- Hand Tools





ROLE:

Mechanics (3D Design, Component Selection, Assembly)

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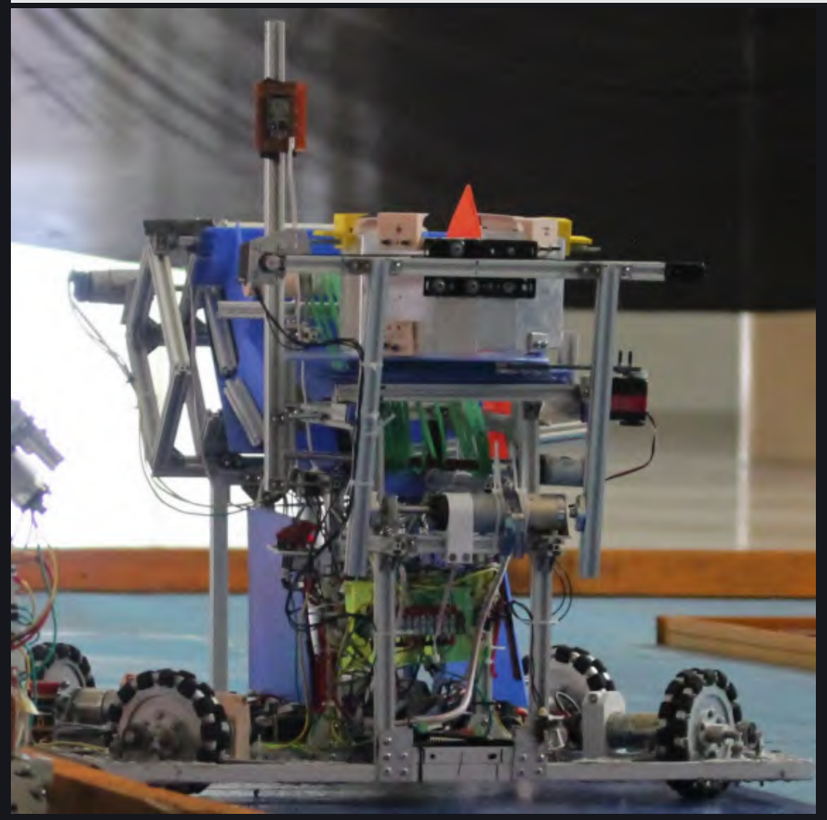
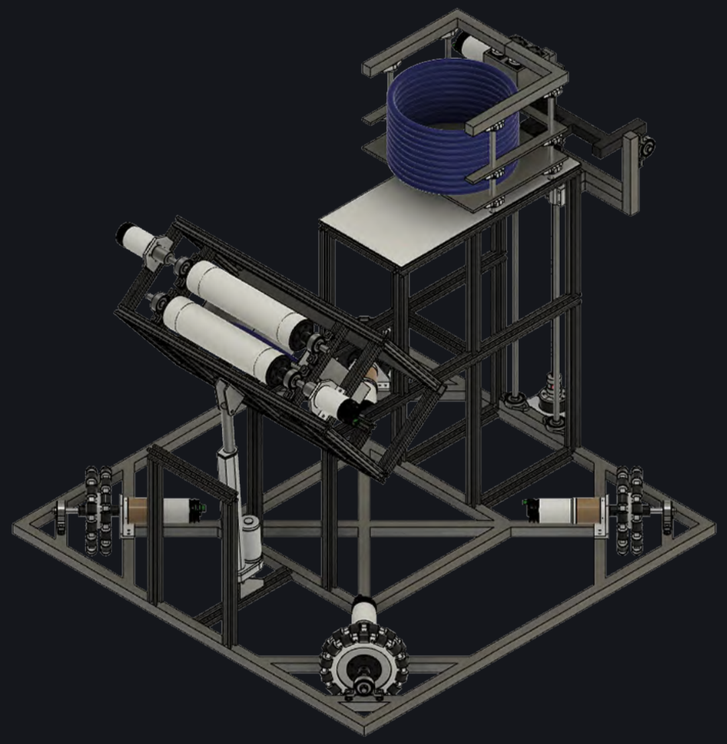
SELECTED PROJECTS

ABU ROBOCON 2023 ROBOT





DESCRIPTION:

A Man-Operated Ring Throwing Robot, capable of picking and throwing rings into poles.

TOOLS & SKILLS:

- 3D CAD
- 3D Printing
- Mechanism Design
- Grinding
- Drilling
- Hand Tools

ROLE:

Mechanics (3D Design, Component Selection, Assembly)

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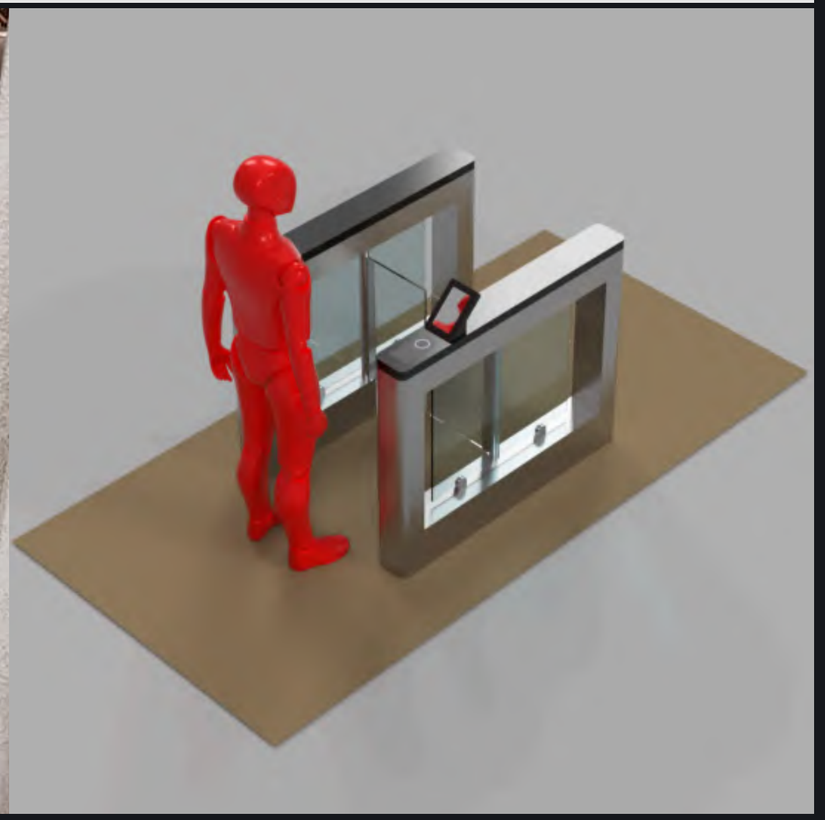
SKYWARD PROJECT - ENTRANCE GATE

DESCRIPTION:

An Entrance Gate Concept and Mounted Display, customized for ZKTeco Gate with a 3D printed display casing.

TOOLS & SKILLS:

- 3D CAD
- 3D Printing
- Product Design
- Product Finishing
- Grinding
- Drilling
- Hand Tools



ROLE:

3D Design Specialist - Mechatronics Engineer

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SELECTED PROJECTS

SKYWARD PROJECT - KID'S GAME

DESCRIPTION:

The Busy Table at Zeppelin, designed for kids' games from acrylics and 3D printed materials.

TOOLS & SKILLS:

- 3D CAD
- 3D Printing
- Mechanism Design
- Product Design
- Product Finishing
- Hand Tools

ROLE:

3D Design Specialist - Mechatronics Engineer

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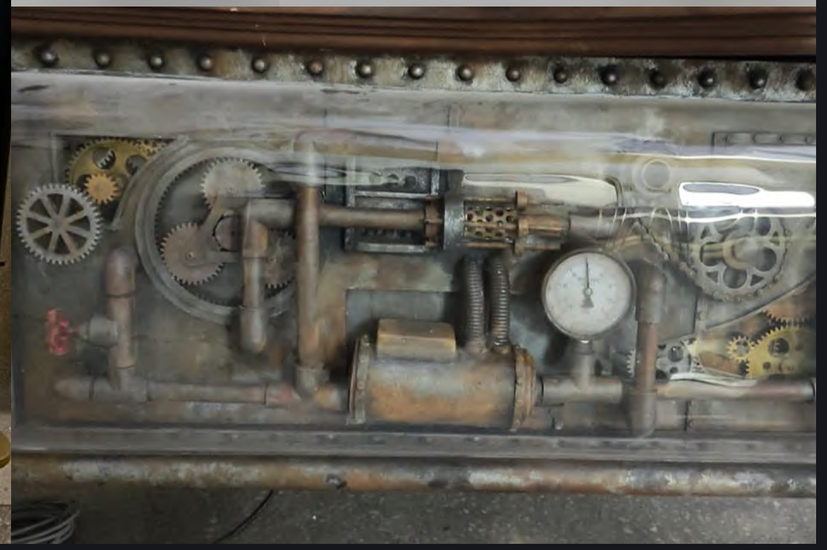
SKYWARD PROJECT - CARD DISPENSER

DESCRIPTION:

A Card Dispensing Machine, including gear sets as the machine's interactive display.

TOOLS & SKILLS:

- 3D CAD
- 3D Printing
- Mechanism Design
- Product Design
- Product Finishing



ROLE:

3D Design Specialist - Mechatronics Engineer

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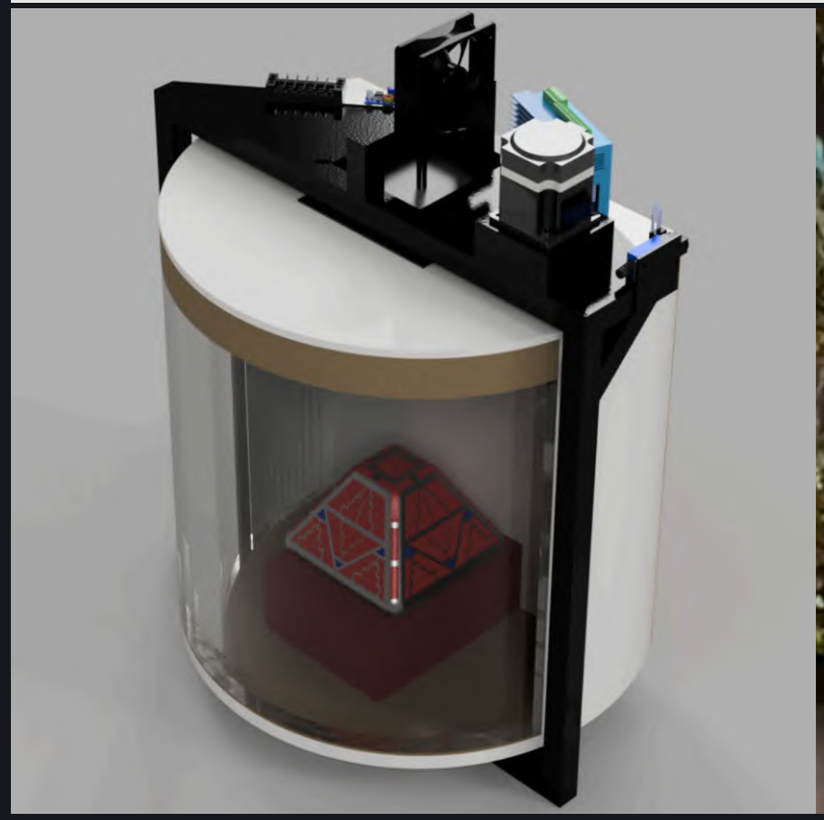
SELECTED PROJECTS

MECHATRONICS DISPLAY AT MUNAS



DESCRIPTION:

An Automatic Artifact Display, using six synchronized stepper motors for a Museum Nasional event.

TOOLS & SKILLS:

- 3D CAD
- Mechanism Design
- Microcontroller
- C++ Programming

ROLE:

Mechatronics Engineer (3D Design & Programming)

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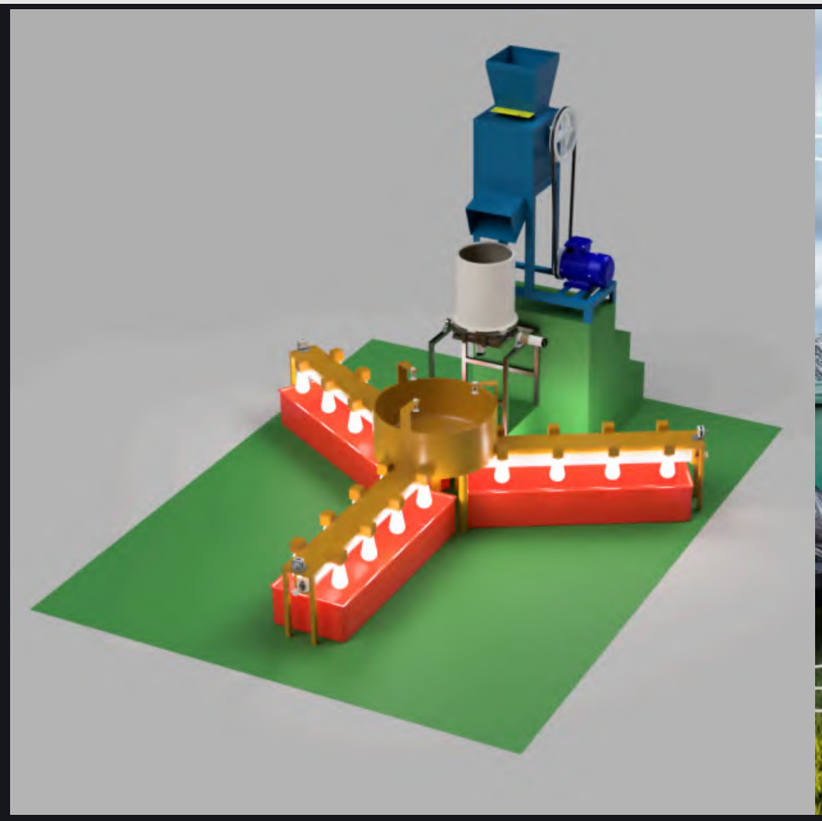
MAGGOT FEEDER FOR INNOVILLAGE CLIENT

DESCRIPTION:

Maggofeed: A Maggot Feeder for an organic waste processing system.

TOOLS & SKILLS:

- 3D CAD
- Mechanism Design
- Product Design
- Detailed Blueprints



INNOVILLAGE

ROLE:

Mechatronics Engineer (3D Design Concept)

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SELECTED PROJECTS

AUTOMATIC CAT FEEDER PROTOTYPE

DESCRIPTION:

An Automatic Cat Feeder, using sensors, a motor, and an ESP32 connected to Kodular and Antares for an IoT project.

TOOLS & SKILLS:

- 3D CAD
- 3D Printing
- Mechanism Design
- Product Design

ROLE:

3D Printing Engineer (3D Design, 3D Printing, Assembly)

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Contacts:

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SELECTED PROJECTS

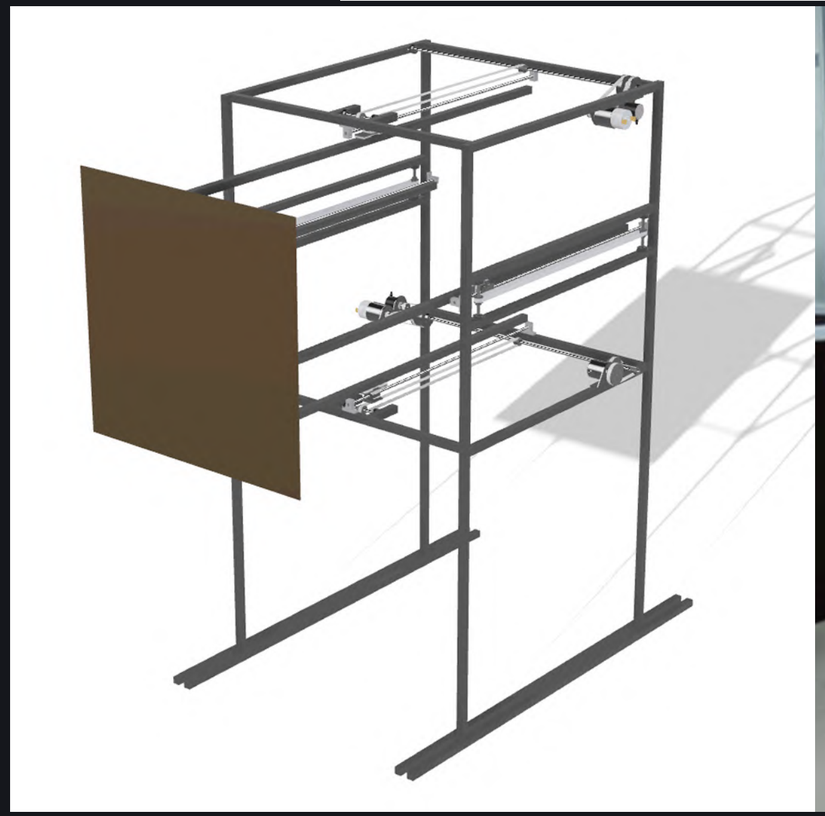
RADAR SLIDER FOR RESEARCH

DESCRIPTION:

An Automatic Mechatronics Slider, a radar/reflector device supporting state-of-the-art radar research.

TOOLS & SKILLS:

- 3D CAD
- Mechanism Design
- Grinding
- Drilling
- Hand Tools



ROLE:

Mechatronics Engineer (3D Design & Assembly)

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SELECTED PROJECTS

1. ABU ROBOCON ROBOTS

- Fully Self-Designed and Built Robots
- Competed at the National Stage of Indonesian Robocon (KRI)
- Full Game Points in Regional Qualification
- 4th Place Finish in the 2024 National Stage

Link: abu-robocon-2024
abu-robocon-2023

2. SKYWARD PROJECT

- 3D Design Specialist at Arthatronic Studio
- Designed and Manufactured for 3D Printing and Acrylics
- Executed Client-Specified Requests
- Contributed to Project Phases 1 & 2

Link: skyward-project

3. MECHATRONICS DISPLAY

- Full Component Selection and Integration
- Designed with Blueprints for Manufacturing
- Programmed and Synchronized Six Motors Seamlessly
- Featured at Munas Repopening Event

Link: mechatronics-artifact-display

4. MAGGOT FEEDER

- Client Project for Innovillage Team
- Full Component Selection for Blueprints
- Developed Design Concept with Blueprints for Manufacturing
- Recognized by Telkom University

Link: maggot-feeder

5. AUTOMATIC CAT FEEDER

- Fully Self-Designed and Built Mechanism
- Incorporated Sensors, Actuators, and Microcontroller Integration
- Optimized Design for 3D Printing
- Achieved A Grade in IoT Class

Link: automatic-cat-feeder

6. RADAR SLIDER

- Supported State-of-The-Art Radar Research
- Fully Self-Designed and Built Mechanism
- Incorporated Sensors, Actuators, and Microcontroller Integration
- Adaptive Design for Radar Size Adjustment

Link: radar-slider

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OTHER PROJECTS

Link: other-projects

SMART WATCH

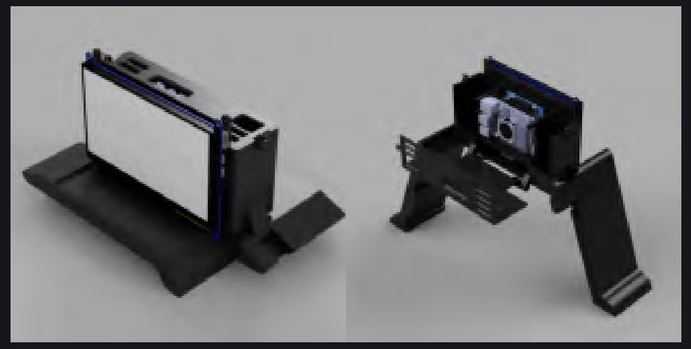
An arrhythmia detection watch.

PANEL BOX

Custom electronic panel boxes with displays.

GRIM REAPER

A 3D printed grim reaper figurine.



PROTABLE CAMERA DEVICE

A vena detection device using raspi camera.

FLEXIBLE KEYCHAINS

Various 3D printed flexible keychains.

LINE FOLLOWER ROBOT

An educational modular robot concept.

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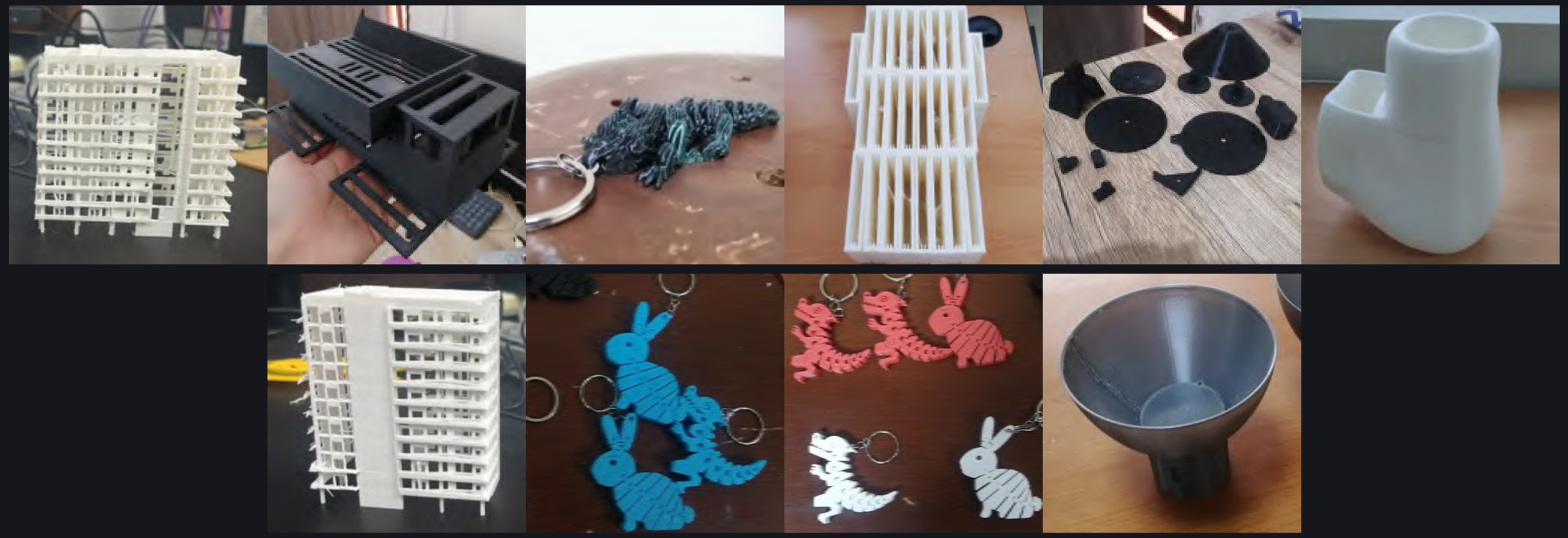
OTHER PROJECTS

Link: other-projects



ACCESSORIES, MINIATURES, TOYS, KEYCHAINS, MOCKUPS, PROTOTYPES, ETC

and more design & printing requests.....

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Contacts:

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CONTACTS & LINKS

More documentations, details, price lists, and free consultations, reach us on:

WhatsApp: wa.me/+6282182227816

Instagram: instagram.com/ariqshaq

LinkedIn: linkedin.com/in/ariqshaq

GitHub: github.com/ariqshaq/3d-design-portfolio

"DESIGNING SOLUTIONS FOR YOUR VISION"

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3D DESIGN & 3D PRINTING SOLUTIONS